

Applications of CAVI: LDA and Poisson Matrix Factorization

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1 Introduction

In this project, we present two different Bayesian hierarchical models whose specifications lend themselves to using Coordinate Ascent Variational Inference (CAVI) for posterior inference. We start by presenting the Latent Dirichlet Allocation (LDA) model, a model that first emerged for text clustering. We present the CAVI coordinate-wise updates before introducing an example dataset and fitting the LDA model to look at the insights we can obtain from posterior inference using the model. We then explore a related model, Poisson Matrix Factorization, which emerged in the 2000s as a model for recommendation algorithms. We will present the model along with the CAVI updates before stating the connections between LDA and Poisson Matrix Factorization.

2 Latent Dirichlet Allocation (LDA)

Latent Dirichlet Allocation (LDA) is an example of a Bayesian hierarchical model using exponential family distributions that allows for the easier computation of CAVI coordinate updates. The model was introduced for text modeling in Blei et al. [1] and for use in genetics from Pritchard et al. [8]. We will start by presenting the model for clustering textual data drawing on Blei et al. [1] and Schein [11]. At a high level, LDA is a mixture model where, unlike something like a Gaussian Mixture Model, we have two different mixtures being modeled. In the context of text data, we will model documents as a mixture of latent topics and each latent topic as a mixture of words from a vocabulary. We will start by specifying the model, deriving the CAVI updates, and writing an expression for the ELBO before implementing these updates using the 20 newsgroups dataset from scikit-learn to see an example of the kinds of insights we can derive from using LDA for posterior inference in the Bayesian setting.

2.1 Description of Model

We will largely use the notation of Schein [11] for describing what is called the “smoothed” LDA model in Blei et al. [1]. We begin with a dataset of documents $w_d, d = 1, \dots, D$ each consisting of N_d words. We write each word

$$w_{dn} \in \{1, \dots, V\}, n = 1, \dots, N_d$$

where we represent English words as numbers in a vocabulary of size V . We then assume that each document is composed of a mixture of K topics parameterized by a matrix $\Theta \in \mathbb{R}^{D \times K}$ where each row of the matrix θ_d corresponds to the mix for document d and must sum to 1.

Separately, we assume each latent topic is composed of a mixture of the V words in the vocabulary, parameterized by the matrix $\Phi \in \mathbb{R}^{K \times V}$ where each row ϕ_k corresponds to the mix for a single topic and must sum to 1.

We then set Dirichlet priors on each θ_d and ϕ_k with

$$\theta_d \sim \text{Dirichlet}(\alpha_1, \dots, \alpha_K)$$

$$\phi_k \sim \text{Dirichlet}(\beta_1, \dots, \beta_V).$$

We will let η represent the hyperparameter vectors of α and β .

Finally, we introduce a latent variable by assuming that for each word in each document, we first draw a topic z_{dn} from the mixture of topics over the document d . We then choose the word w_{dn} from the mixture of words over topic z_{dn} . That is,

$$z_{dn} \sim \text{Categorical}(\theta_d)$$

$$w_{dn} \sim \text{Categorical}(\phi_{z_{dn}}).$$

We also note that the support of the discrete categorical distribution is given by an integer value; so for example, the support of $\text{Categorical}(\theta_d)$ is $\{1, \dots, K\}$ which we use to identify the mix in the second stage of the generative process.

So, collecting all documents into W and latent z_{dn} into Z , we have joint density of the model given by

$$\begin{aligned} p(W, Z, \Theta, \Phi \mid \eta) &= p(\Theta \mid \eta) p(\Phi \mid \eta) p(Z \mid \Theta) p(W \mid Z, \Phi) \\ &= \left(\prod_{d=1}^D \prod_{k=1}^K \theta_{dk}^{\alpha_k - 1} \right) \left(\prod_{k=1}^K \prod_{v=1}^V \phi_{kv}^{\beta_v - 1} \right) \left(\prod_{d=1}^D \prod_{n=1}^{N_d} \prod_{k=1}^K [\theta_{dk}]^{\mathbb{1}\{z_{dn}=k\}} \right) \left(\prod_{d=1}^D \prod_{n=1}^{N_d} \prod_{k=1}^K \prod_{v=1}^V [\phi_{kv}]^{\mathbb{1}\{w_{dn}=v, z_{dn}=k\}} \right). \end{aligned}$$

We note the indicator functions in the exponents which allow us to write the joint density as a product over k for $p(Z \mid \Theta)$ and products over k, v for $p(W \mid Z, \Phi)$.

2.2 CAVI Updates

We now wish to do inference on the posterior $p(\Theta, \Phi, Z \mid W, \eta)$ which will be our target distribution for the CAVI algorithm. We will refer to the group of Θ, Φ , and Z as our latent variables moving forward. We will then try to find the variational distribution

$$g(\Theta, \Phi, Z) = \prod_{d=1}^D g(\theta_d) \prod_{k=1}^K g(\phi_k) \prod_{d=1}^D \prod_{n=1}^{N_d} g(z_{dn})$$

to approximate the posterior.

As noted in Remark 10.8, the CAVI updates are proportional to the exponential of the expectation of the log of the full conditional. So to get the CAVI updates, we can first derive the full conditionals for our latent variables then compute the coordinate-wise updates. Since the probability distributions we're working with are in the exponential family, our computations become easier.

2.2.1 Full Conditionals

We start with the full conditionals for Θ and Φ which will be similar. By the definition of conditional probability, we know that

$$p(\Theta \mid Z, W, \Phi, \eta) \propto p(W, Z, \Theta, \Phi \mid \eta).$$

So to derive the complete conditional, we can just pull all terms from the joint density that depend on Θ . We write

$$p(\theta_d \mid \Phi, Z, W, \eta) \propto \prod_{k=1}^K \theta_{dk}^{\alpha_k - 1} \prod_{n=1}^{N_d} \prod_{k=1}^K \theta_{dk}^{\mathbb{1}\{z_{dn}=k\}}$$

$$= \prod_{k=1}^K \theta_{dk}^{\alpha_k + \sum_{n=1}^{N_d} \mathbb{1}\{z_{dn}=k\} - 1} \implies p(\theta_d \mid \Phi, Z, W, \eta) = \text{Dirichlet}(\alpha_1 + y_{d1}, \dots, \alpha_K + y_{dK})$$

where

$$y_{dk} = \sum_{n=1}^{N_d} \mathbb{1}\{z_{dn} = k\}.$$

So the full conditional requires updating the prior parameters by the count of words where topic k was chosen in the document. We get a similar style of update for ϕ_k with

$$p(\phi_k \mid \Theta, Z, W, \eta) = \text{Dirichlet}(\beta_1 + y_{k1}, \dots, \beta_V + y_{kV})$$

$$y_{kv} = \sum_{d=1}^D \sum_{n=1}^{N_d} \mathbb{1}\{z_{dn} = k, w_{dn} = v\}.$$

So for the ϕ_k , which corresponds to the mixture for topic k , we update prior parameters using the count of words across all documents assigned to a given word for that topic.

Then finally, we can derive the full conditional for z_{dn} . We can write

$$p(z_{dn} = k \mid w_{dn} = v, \Theta, \Phi, \eta) \propto \theta_{dk} \phi_{kv} \implies p(z_{dn} = k \mid w_{dn} = v, \Theta, \Phi, \eta) = \frac{\theta_{dk} \phi_{kv}}{\sum_{j=1}^K \theta_{dj} \phi_{jv}}$$

where we get the simplified terms due to the way the indicator functions simplify in the joint distribution. This corresponds to a categorical distribution again with probabilities for each k given above.

2.2.2 Coordinate Updates

We can then use update rule equivalence from Remark 10.8 in the lecture notes

$$g_i^*(x_i) \propto \exp(\mathbb{E}_{g_{-i}}[\log f(x_i \mid x_{-i})])$$

to derive the coordinate updates for the model. We begin with $g(\Theta)$, and we will start by using $\mathbb{E}_{g_{-\Theta}}$ to indicate the expectation over $g(\Phi, Z)$. We can then write, keeping only terms that include θ_{dk} ,

$$\begin{aligned} g^*(\theta_d) &\propto \exp(\mathbb{E}_{g_{-\Theta}}[\log \text{Dirichlet}(\alpha_1 + y_{d1}, \dots, \alpha_K + y_{dK})]) \\ &\propto \exp\left(\sum_{k=1}^K \log(\theta_{dk}) \mathbb{E}_{g_{-\Theta}}[\alpha_k + y_{dk} - 1]\right) \propto \prod_{k=1}^K \theta_{dk}^{\alpha_k + \mathbb{E}_{g_{-\Theta}}[y_{dk}] - 1}. \end{aligned}$$

So we see that the coordinate-wise update for θ_d is also a Dirichlet, and it requires computing the expectation

$$\mathbb{E}_{g_{-\Theta}}[y_{dk}] = \sum_{n=1}^{N_d} \mathbb{E}_{g_{-\Theta}}[\mathbb{1}\{z_{dn} = k\}] = \sum_{n=1}^{N_d} g(z_{dn} = k).$$

So in order to compute the update for θ_d , we need to use the variational distribution for z_{dn} . Since ϕ_k has a full conditional that is also a Dirichlet, we similarly get a coordinate-wise update that is Dirichlet with

$$g^*(\phi_k) = \text{Dirichlet}(\beta_1 + \mathbb{E}_{g_{-\Phi}}[y_{k1}], \dots, \beta_V + \mathbb{E}_{g_{-\Phi}}[y_{kV}])$$

so we will need to compute each round

$$\mathbb{E}_{g_{-\Phi}}[y_{kv}] = \sum_{d=1}^D \sum_{n=1}^{N_d} \mathbb{1}\{w_{dn} = v\} g(z_{dn} = k).$$

We note that for the coordinate updates, we treat the words w_{dn} as fixed, so the indicator in the above expectation remains for w_{dn} .

Finally, we can derive the update for $g(z_{dn})$ as follows

$$\begin{aligned} g^*(z_{dn} = k) &\propto \exp(\mathbb{E}_{g_{-Z}}[\log \theta_{dk} + \log \phi_{kv}]) \\ &\propto \exp(\mathbb{E}_{g_{-Z}}[\log \theta_{dk}]) \exp(\mathbb{E}_{g_{-Z}}[\log \phi_{kv}]). \end{aligned}$$

In this derivation, we've dropped the denominator from the full conditional since this does not depend on the choice of z_{dn} . We see that this coordinate-wise update is similar to the full conditional for z_{dn} and will require calculating the expectations of the logs of Dirichlet random variables θ_{dk} and ϕ_{kv} . We will take for granted that $\mathbb{E}[\log X_i]$ for $X = (X_1, \dots, X_n) \sim \text{Dirichlet}(\alpha_1, \dots, \alpha_n)$ is given by

$$\mathbb{E}[\log X_i] = \psi(\alpha_i) - \psi\left(\sum_{i=1}^n \alpha_i\right)$$

where ψ is the digamma function.

So we now have all the ingredients required to compute the coordinate-wise updates for implementing CAVI for an example. Before the implementation, we will derive an expression for the ELBO that we can use to monitor the performance of the algorithm over iterations.

2.2.3 Expression for ELBO(g)

We now sketch an expression for the ELBO that we will compute after every iteration of CAVI to observe the convergence of the algorithm. As our target is the posterior, we have our “unnormalized” version as the joint density from Section 2.1, which incorporates the likelihood of the model and the two Dirichlet priors we set. As in the lecture notes, we will start by labeling this distribution as $\tilde{f}$ but will then see that decomposing it into pieces will make it more tractable. We will also write for the variational distribution the shorthand

$$g(\Theta) = \prod_{d=1}^D g(\theta_d)$$

and likewise for Φ and Z .

We can write

$$\begin{aligned}
\text{ELBO}(g) &= \mathbb{E}_g[\log \tilde{f}] - \mathbb{E}_g[\log g] \\
&= \mathbb{E}_g[\log(p(W | Z, \Phi)p(Z | \Theta)p(\Phi)p(\Theta))] - \mathbb{E}_g[\log(g(\Theta)g(\Phi)g(Z))] \\
&= \mathbb{E}_g\left[\log\left(\frac{p(W | Z, \Phi)p(Z | \Theta)}{g(Z)}\right)\right] + \mathbb{E}_g\left[\log\left(\frac{p(\Phi)}{g(\Phi)}\right)\right] + \mathbb{E}_g\left[\log\left(\frac{p(\Theta)}{g(\Theta)}\right)\right] \\
&= \mathbb{E}_g\left[\log\left(\frac{p(W | Z, \Phi)p(Z | \Theta)}{g(Z)}\right)\right] - d_{KL}(g(\Phi) \parallel p(\Phi)) - d_{KL}(g(\Theta) \parallel p(\Theta)).
\end{aligned}$$

We refrain from writing out the precise terms for each calculation, but this is the factorization used in the implementation of the ELBO used for the subsequent example. This factorization is inspired by the fact that pytorch provides a function to compute the KL divergence between two distributions.

2.3 Example Application

We now move to implementing the CAVI updates for the 20 newsgroups dataset from scikit-learn. The dataset we use is comprised of 5,000 documents that come from 20 different news desks which act as categories. We can use these 20 news desk classifications as something to compare with the latent topics we discover from our LDA model to see if the model discovers patterns that reflect the different news desks. Before implementing CAVI updates, we do some text cleaning to make the task more tractable and interpretable. We keep just the nouns, adjectives, verbs, and adverbs by parsing document text using the spacy library. We then drop words that appear in either more than 50% of documents or fewer than 10 documents to eliminate words in all documents and edge cases.

After doing this data preparation, we have a vocabulary of size $V = 3413$ for $D = 5000$ documents. We then assume there are $K = 20$ latent topics and run CAVI for 200 iterations. Figure 1 shows the increase in the ELBO by iteration (averaged by total word count), showing the flattening trend in later iterations.

2.3.1 Inference

With our variational distribution, we can see what latent topics have been discovered within the text and how this model formulation can provide practical use. The first thing we can look at is the top words associated with each topic to see if the recovered topics have any semantic interpretation. We present word clouds for two of the more distinctive topics in Figure 2; the word cloud for topic 6 appears to correspond to sports related articles and topic 12 appears to correspond to space related articles.

We can then compare the latent topics with the 20 newsdesk classifications for the original documents. We can examine the variational distribution $g(\Theta)$ to gain insights on the learned mixtures for individual documents. Two example classifications among the 20 provided are “rec.sport.hockey” and “sci.space”; we would expect that articles from these newsdesks would probably have a mixture with more weight corresponding to the topics 6 and 12 we looked at above. We find that this holds, with almost 70% of the 261 “rec.sport.hockey” articles having topic 6 as the largest component of the



Figure 1 ELBO Value

mixture and a plurality 46% of 248 “sci.space” articles having topic 12 as the most prominent mixture component.

These are two simple examples of how we can use the variational distributions to discover semantic meaning in text data. Although we have the prior classifications of documents, these are not used in fitting the model and are only used in the analysis post-hoc. One could imagine other text corpora with supplementary data for each document that would allow the topics to be used as tools for characterizing groups of documents in different ways.

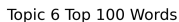


Figure 2 Top 100 words in word clouds for two topics

2.4 Other Applications and Ideas

We now briefly review the second foundational LDA paper, applying the model using genotype data before looking at some papers that have used LDA for text modeling within psychology and healthcare.

Alongside the application of LDA for text modeling, Pritchard et al. [8] introduced LDA to discover latent “populations” where each document is instead a human. Instead of words, each individual is composed of a set of alleles and the authors use the term “admixture” to refer to an individual being a mix of different populations. The authors present examples using bird and human genotype data where the discovered LDA populations correspond to the physical geographies of origins for the individuals.

Chiu et al. [2] used LDA to discover latent topics within the descriptions of negative self-images provided by a sample of 11-15 year-old students with the goal of identifying themes across responses. The authors discovered two topics with one topic characterized by words describing visible signs of anxiety and the other characterized by feelings of judgment. The authors also systematically chose to model two topics by assessing topic coherence for different choices of K topics. We note that in the example above, we chose $K = 20$ as it corresponded to the underlying topics from the dataset, although other choices might lead to higher topic coherence. Topic coherence can be measured in different ways; one way involves comparing the top words for the latent topics with word occurrences within a large corpus of text data like Wikipedia. Another way that does not involve using external data involves comparing topic words with word occurrences within the documents being used. Both these methods are described in further detail in Röder et al. [9] along with a unifying framework for both these ideas.

LDA is often paired with some other observed variables for a text dataset so that the latent topics can be used as an additional angle with which to view the dataset. For example, Feldhege et al. [3] uses LDA to identify topics from discussion posts from the online forum r/depression on the social media platform Reddit. The authors use LDA to find 26 interpretable topics among discussion posts, which they use to answer questions like which topics correspond to posts with more engagement measured by things like the number of comments on a post. In another example, Geletta et al. [5] uses LDA on the descriptions of clinical trials in an effort to identify themes in terminated clinical studies.

3 Poisson Matrix Factorization

We now discuss another model where CAVI can be used for inference, another hierarchical model using exponential family distributions in a way that simplifies the coordinate-wise updates. Poisson Matrix Factorization emerged as a useful method for developing recommendations from user data with either implicit integer feedback (i.e. number of watches/streams) or explicit integer feedback (i.e. 1-5 stars). The specification of the model has a similar structure to that of LDA, and we will note a direct equivalence between Poisson Matrix Factorization and LDA if we specify the model in a specific way.

3.1 Description of Model

Suppose we have data in the format $X \in \mathbb{Z}_{\geq 0}^{N \times M}$ where each row corresponds to an individual customer, each column corresponds to a product, and each entry in X represents the number of times the customer has bought the product. We assume that each entry in the matrix is a nonnegative integer, so we can model each x_{ij} as a Poisson random variable. We assume that among the products, there are K latent product categories and wish to “factor” X into

$$X \approx \Theta \Phi, \Theta \in \mathbb{R}_+^{N \times K}, \Phi \in \mathbb{R}_+^{K \times M}.$$

We can imagine the matrix Θ as modeling how much each customer is associated with each product category and the matrix Φ as how we associate each product with a product category. Let $\theta_{i\bullet}^T$ represent the i th row of Θ and $\phi_{\bullet j}$ represent the j th column of Φ . Then we assume

$$x_{ij} \sim \text{Poisson}(\theta_{i\bullet}^T \phi_{\bullet j}) = \text{Poisson}\left(\sum_{k=1}^K \theta_{ik} \phi_{kj}\right).$$

We will also choose to put Gamma priors over θ_{ik}, ϕ_{kj} as follows

$$\theta_{ik} \sim \text{Gamma}(\alpha_\theta, \beta_\theta)$$

$$\phi_{kj} \sim \text{Gamma}(\alpha_\phi, \beta_\phi)$$

$$\eta = \{\alpha_\theta, \beta_\theta, \alpha_\phi, \beta_\phi\}$$

One could then imagine doing inference over the posteriors for Θ, Φ as a method of understanding customer preferences and recommending similar products. We can again use the full conditionals to specify the CAVI updates, however, in this case, the full conditional $p(\theta_{ik} \mid X, \Phi, \eta)$ is intractable. To deal with this and make the full conditionals tractable, we introduce auxiliary variables into the model.

3.1.1 Introducing Auxiliary Variables

Using the fact that the sum of independent Poisson random variables is a Poisson random variable with rate given by the sum of the rates, we can write

$$x_{ij} = \sum_{k=1}^K z_{ijk}, z_{ijk} \sim \text{Poisson}(\theta_{ik} \phi_{kj})$$

which is equivalent to our original model assuming each z_{ijk} are drawn independently. Grouping all z_{ijk} into $Z \in \mathbb{R}^{N \times M \times K}$, we can write the joint distribution of the model (as given in Schein [10]) as

$$\begin{aligned} p(X, Z, \Theta, \Phi \mid \eta) = & \left[\prod_{i=1}^N \prod_{j=1}^M \mathbb{1} \left\{ x_{ij} = \sum_{k=1}^K z_{ijk} \right\} \prod_{k=1}^K \text{Poisson}(z_{ijk} \mid \theta_{ik} \phi_{kj}) \right] \\ & \times \left[\prod_{i=1}^N \prod_{k=1}^K \text{Gamma}(\theta_{ik} \mid \alpha_\theta, \beta_\theta) \right] \left[\prod_{j=1}^M \prod_{k=1}^K \text{Gamma}(\phi_{kj} \mid \alpha_\phi, \beta_\phi) \right]. \end{aligned}$$

From this joint distribution, we can derive full conditionals and then the CAVI updates in a similar manner to the LDA derivations.

3.2 Coordinate Updates

We now proceed through a more abbreviated derivation of the CAVI updates for the Poisson Matrix Factorization model. As before, our target for inference is the posterior $p(\Theta, \Phi, Z \mid X, \eta)$.

Similar to the derivations of the LDA model, the use of exponential family distributions leads to tractable full conditionals but only since we introduced the auxiliary variables. Just as in the LDA model with the Dirichlet priors leading to Dirichlet coordinate updates in the CAVI algorithm, we find similar updates using the Gamma priors for Θ and Φ . For θ_{ik} , we can use the joint distribution above to derive the full conditional

$$f(\theta_{ik} \mid \Phi, Z, X, \eta) \propto \theta_{ik}^{\alpha_\theta - 1} \exp(-\beta_\theta \theta_{ik}) \prod_{j=1}^M (\theta_{ik} \phi_{kj})^{z_{ijk}} \exp(-\theta_{ik} \phi_{kj})$$

$$\implies \theta_{ik} \mid \Phi, Z, X, \eta \sim \text{Gamma}\left(\alpha_\theta + \sum_{j=1}^M z_{ijk}, \beta_\theta + \sum_{j=1}^M \phi_{kj}\right).$$

Then the CAVI update for $g(\theta_{ik})$ requires computing the expectations

$$\mathbb{E}_{g-\Theta} \left[\sum_{j=1}^M z_{ijk} \right], \mathbb{E}_{g-\Theta} \left[\sum_{j=1}^M \phi_{kj} \right]$$

with analogous updates for $g(\phi_{kj})$

$$\mathbb{E}_{g-\Phi} \left[\sum_{i=1}^N z_{ijk} \right], \mathbb{E}_{g-\Phi} \left[\sum_{i=1}^N \theta_{ik} \right].$$

Moving to the full conditional of $z_{ij\bullet} = (z_{ij1}, \dots, z_{ijk})^T$, we note that in the full conditional for z_{ij} , we condition on the sum of the vector being fixed at x_{ij} . Then the joint distribution of z_{ij} is Multinomial with count x_{ij} and probabilities $p_k \propto \theta_{ik} \phi_{kj}$. The CAVI updates for each z_{ij} requires keeping track of the expectations

$$\mathbb{E}_{g-Z} [\log \theta_{ik}], \mathbb{E}_{g-Z} [\log \phi_{kj}].$$

The expected value of the log of a Gamma-distributed random variable is calculable in closed form, so each of these CAVI updates is easily calculable for implementation.

3.3 Applications

Authors have applied this model in different settings and we present two examples here. In Liang et al. [7], the authors consider the task of assigning labels to individual songs, with each row of the data corresponding to a song and each column corresponding to an integer feature about the song. The authors then can use the variational distributions

fitted using variational inference to either assign features to songs or to retrieve songs most related to a given feature.

Alternatively, Gopalan et al. [6] focuses on Poisson Matrix Factorization for providing recommendations from user feedback. The authors frame the variational distribution as a method of providing recommendations to users, with multiple examples including 1-5 star ratings from Netflix users for movies on the platform. The authors find that this model, which they call in the paper “hierarchical Poisson matrix factorization” outperforms other recommendation models across four different datasets spanning different scenarios.

3.4 Connection to LDA

It is also worth noting that specifying the Poisson Matrix Factorization model in a different way leads to an equivalent parameterization of LDA as found in Geiger and Park [4]. We start with the same matrices $X \in \mathbb{R}^{N \times M}$, $\Theta \in \mathbb{R}^{N \times K}$ and $\Phi \in \mathbb{R}^{K \times M}$ and we continue to assume the same Poisson model for each x_{ij} . The authors then restrict Φ such that each row $\phi_{k\bullet}$ is on the simplex and set a prior on each row of Θ

$$\theta_{d\bullet} \sim \text{Dirichlet}(\tilde{\alpha}).$$

They then show that the ELBO derived using these constraints from the perspective of Poisson Matrix Factorization matches the ELBO obtained from an LDA formulation with only a prior on the document-topic mixes Θ (whereas the model presented here placed a prior on the word-topic mixes Φ as well).

4 Conclusion

Both LDA and Poisson Matrix Factorization present modern examples of cases where CAVI updates are tractable and provide utility for performing inference in applied settings. This paper has looked at both models and implemented CAVI for LDA, although it is worth noting that modern implementations of the LDA model, in the python software scikit-learn for example, often forgo CAVI and instead use stochastic variational inference which scales better to larger datasets.

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